

main.c



Share

Run

```
1 #include <stdio.h>
2 #include <pthread.h>
3 #include <unistd.h>
4
5 #define N 5
6
7 pthread_mutex_t forks[N];
8
9 void* philosopher(void* num) {
10     int id = *(int*)num;
11     while(1) {
12         printf("Philosopher %d thinking\n", id);
13         sleep(1);
14
15         pthread_mutex_lock(&forks[id]);
16         pthread_mutex_lock(&forks[(id+1)%N]);
17
18         printf("Philosopher %d eating\n", id);
19         sleep(1);
20
21         pthread_mutex_unlock(&forks[id]);
22         pthread_mutex_unlock(&forks[(id+1)%N]);
23     }
24 }
25
26 int main() {
27     pthread_t p[N];
28     int i, ids[N];
29
30     for(i = 0; i < N; i++)
31         pthread_mutex_init(&forks[i], NULL);
32
33     for(i = 0; i < N; i++) {
34         ids[i] = i;
35         pthread_create(&p[i], NULL, philosopher, &ids[i]);
36     }
37     for(i = 0; i < N; i++)
38         pthread_join(p[i], NULL);
39     return 0;
40 }
```

Output

```
Philosopher 1 thinking
Philosopher 0 thinking
Philosopher 4 thinking
Philosopher 3 thinking
Philosopher 2 thinking
Philosopher 1 eating
Philosopher 4 eating
Philosopher 1 thinking
Philosopher 4 thinking
Philosopher 3 eating
Philosopher 0 eating
Philosopher 3 thinking
Philosopher 0 thinking
Philosopher 2 eating
Philosopher 4 eating
Philosopher 2 thinking
Philosopher 4 thinking
Philosopher 1 eating
Philosopher 3 eating
Philosopher 3 thinking
Philosopher 1 thinking
Philosopher 0 eating
Philosopher 2 eating
Philosopher 0 thinking
Philosopher 2 thinking
Philosopher 1 eating
Philosopher 4 eating
Philosopher 1 thinking
Philosopher 4 thinking
Philosopher 3 eating
Philosopher 0 eating
Philosopher 3 thinking
Philosopher 0 thinking
Philosopher 4 eating
Philosopher 2 eating
Philosopher 4 thinking
Philosopher 2 thinking
Philosopher 1 eating
Philosopher 3 eating
Philosopher 1 thinking
Philosopher 0 eating
Philosopher 3 thinking
```